Network Protocols

IP :

IPv4

xxxx. xxxx. xxxx. xxxx.

- made up of netID and Host ID
- device within the same net ID is consider on the same subnet, eg. all device with this blue color on diagram on right

IPv6

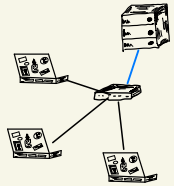
xxxx : xxxx : xxxx : xxxx : xxxx : xxxx : xxxx : xxxx

URL eg. HTTPS // www. google. com

protocol, subdomain, Top level domain, domain name

accessing websites

IP address corresponding to URL



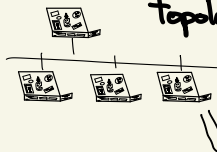
Client server

Thin client : depend heavily on server

Thick client : client can still perform some task when no connect to server

peer to peer

aka P2P

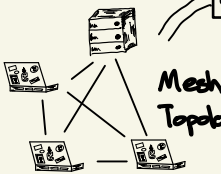


Bus topology

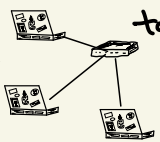
* uses CSMA/CD

Hybrid Topology

combination

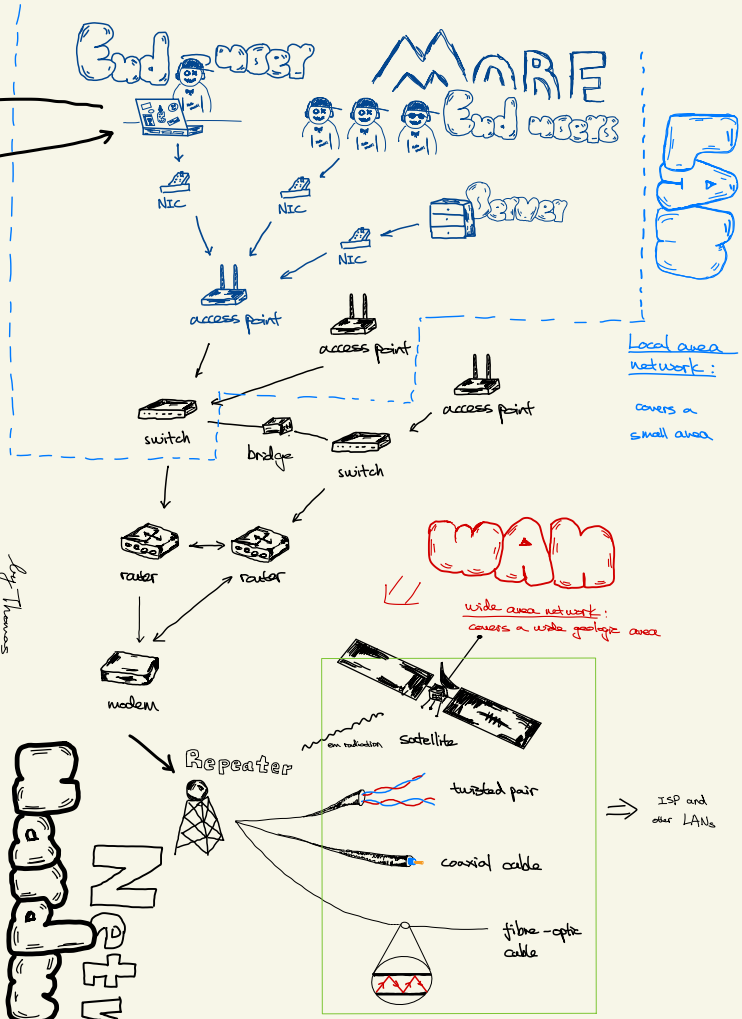


Mesh Topology



Star topology

Network Components



Ways of data transmision

* computer may use architectures such as PSTN to connect

by Thomas

Communication and networking technologies

Derek Huang

WAN - wide area network
LAN - local area network

client - server

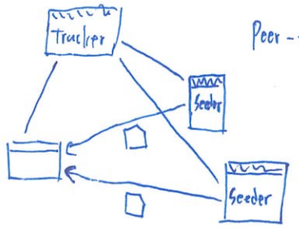
- Alternatively, download application from server and runs it.
- Less latency
- Can operate (full or part of it's functionality) without internet

- Requires internet connection
- Take less to set up a client
- Less cost on the whole setup resources can be reallocated (as most resource is on server)

File Sharing

Client - Server

- Files stored on server, download from server.
- If server is gone, all file gone
- Server may use or even abuse your data (privacy issue)
- Generally secure if the server operator is reliable.
- Download / upload speed may be limited.



Peer - to - peer (P2P)

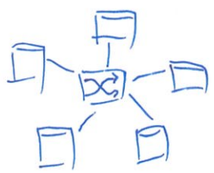
- Decentralized
- Higher privacy
- Lower security (Short board effect)
- Higher possible download speed, parts of a file can be downloaded separately
- Requires seeders, resources can be not available or permanently lost.
- Need public IP to seed, limiting the number of seeders (especially in China)
- May be illegal in some country.
- Everyone can know what you downloads, VPN or proxy can be used to hide ip address.

Network topologies

Bus topology

Mesh topology

Star topology



□ → end systems

Hybrid topology - Mixed topology, depends on actual cases.
↳ · Higher flexibility, but higher cost.

Transmission media

wired (Cable) ———— Wireless - - - - -

	Twisted pair	Coaxial	Fibre-optic	Radio 3kHz – 3GHz	Microwave 3GHz – 300GHz	Infrared 300GHz – 400THz
Cost	→	→	→			
Bandwidth	→	→	→	Bandwidth	→	→
Attenuation at HF	Mid	Most	Least	Attenuation (rain)	→	→
Interference	Most	Mid	Least	Need for repeaters	→	→
Need for repeaters	Often	Often	Less often	Directional focusing	→	→
				Penetration	←	←

Hardwares

Router

Switch

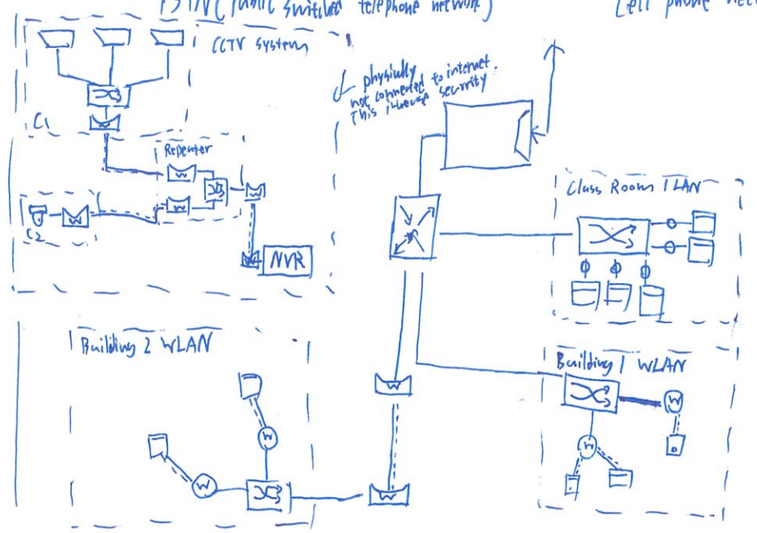
Bridge

Access Point (AP)

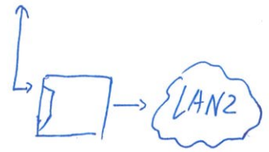
Wireless AP

MoDem (Modulator & Demodulator)

PSTN (Public Switched telephone network)



Cell phone network



Cloud Computing

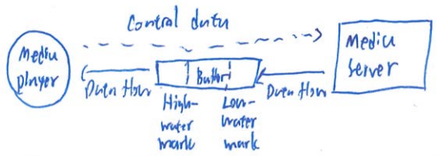
private cloud
- Organisation is responsible
- Org take full cost of creation & maintenance

public cloud
- Use third-party provided, created, managed, owned cloud service.
  1. Infrastructure
  2. Platform
  3. Software

Possible ways of private cloud

1. Org take full responsibility for creating, managing the cloud ~~on site~~ installed on-site and connected to private network
2. outsource to a third-party the creation and management of on-site installation connected to private network
3. outsource the creation and management to third-party of an Internet accessible system.

Bit streaming



IP Addressing

IPv4

0.0.0.0 ~ 255.255.255.255

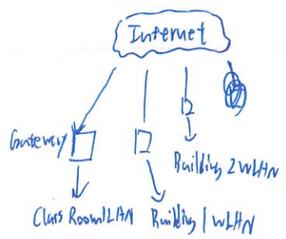
IPv6

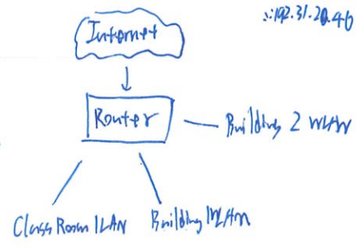
~~0000:0000:0000:0000:0000:0000~~ ~
~~FFFF:FFFF:FFFF:FFFF:FFFF:FFFF~~

:: 6C48: PFFF: ABCD: PCBE =
0000:0000: 6C48: PFFF: ABCD: PCBE

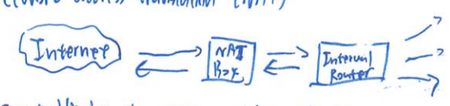
::192.31.20.46 → An IPv4 address used in IPv6.

Sub - netting





Network address translation (NAT)



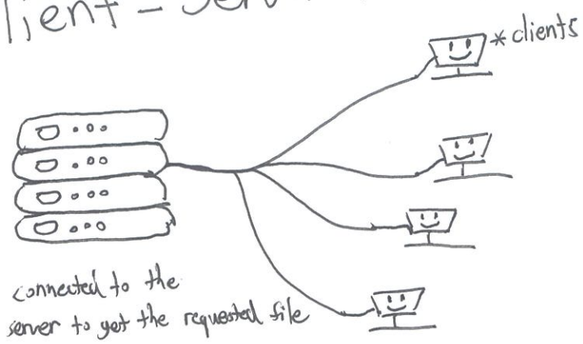
One public ip the NAT viewed by public internet. Inside the private network, private network is often to ensure each ip is unique.

Static & dynamic IP

Static IP won't change.
Dynamic IP is allocated dynamically to increase max user provided above the ip exist. But user at same time is still limited.

Harry

Client - Server Model



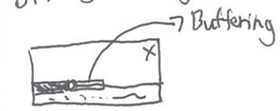
Thin Client Software

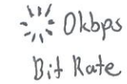
Depends heavily on having access to a server.

Thick Client Software

Doesn't care if it's connected to a server or not

Bit - Streaming





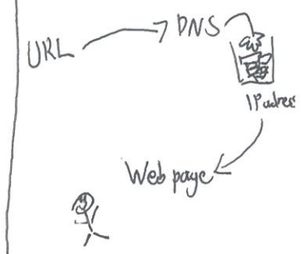
On-Demand

Real-Time

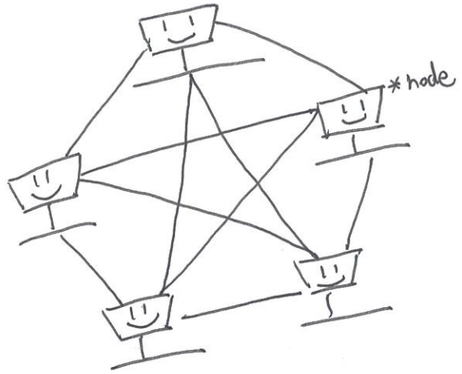
IP Address

IPv4: 192.168....

IPv6: F2F2:FFF1...



Peer - to - Peer Model (P2P)



no central server

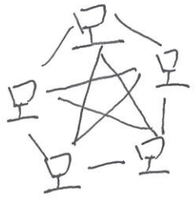
Each nodes can share its files with all other nodes

Bus Topology

contains one shared link to which all devices are attached

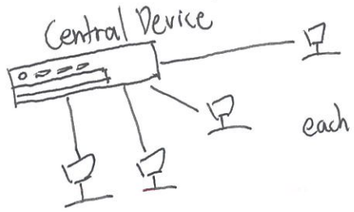
Uses CSMA/CD to avoid collisions

Mesh Topology



contains direct links between devices

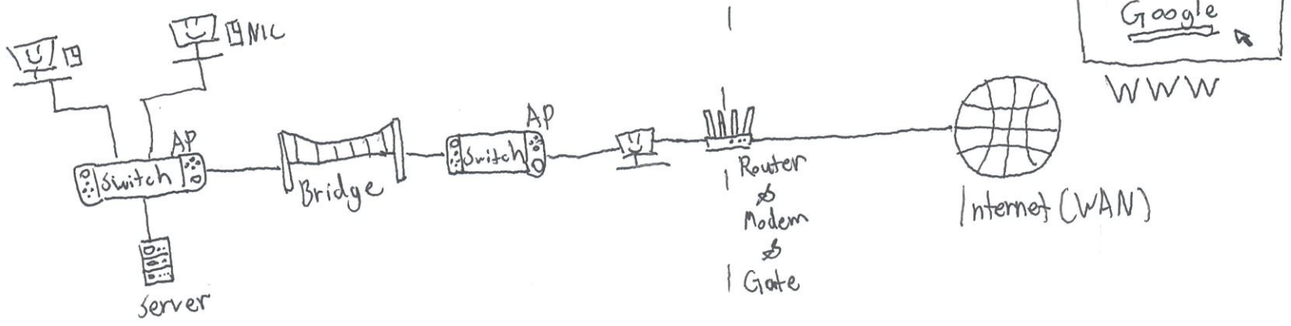
Star Topology



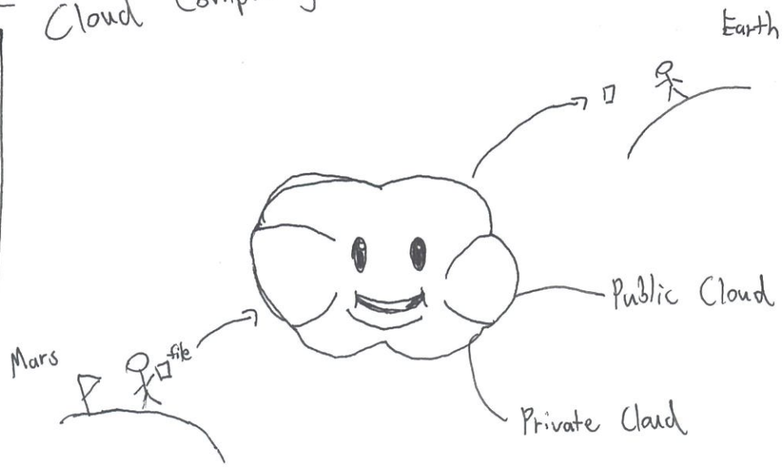
each end-systems is linked to a central device

Hybrid Topology: a collection of connected LANs with different topologies.

LAN

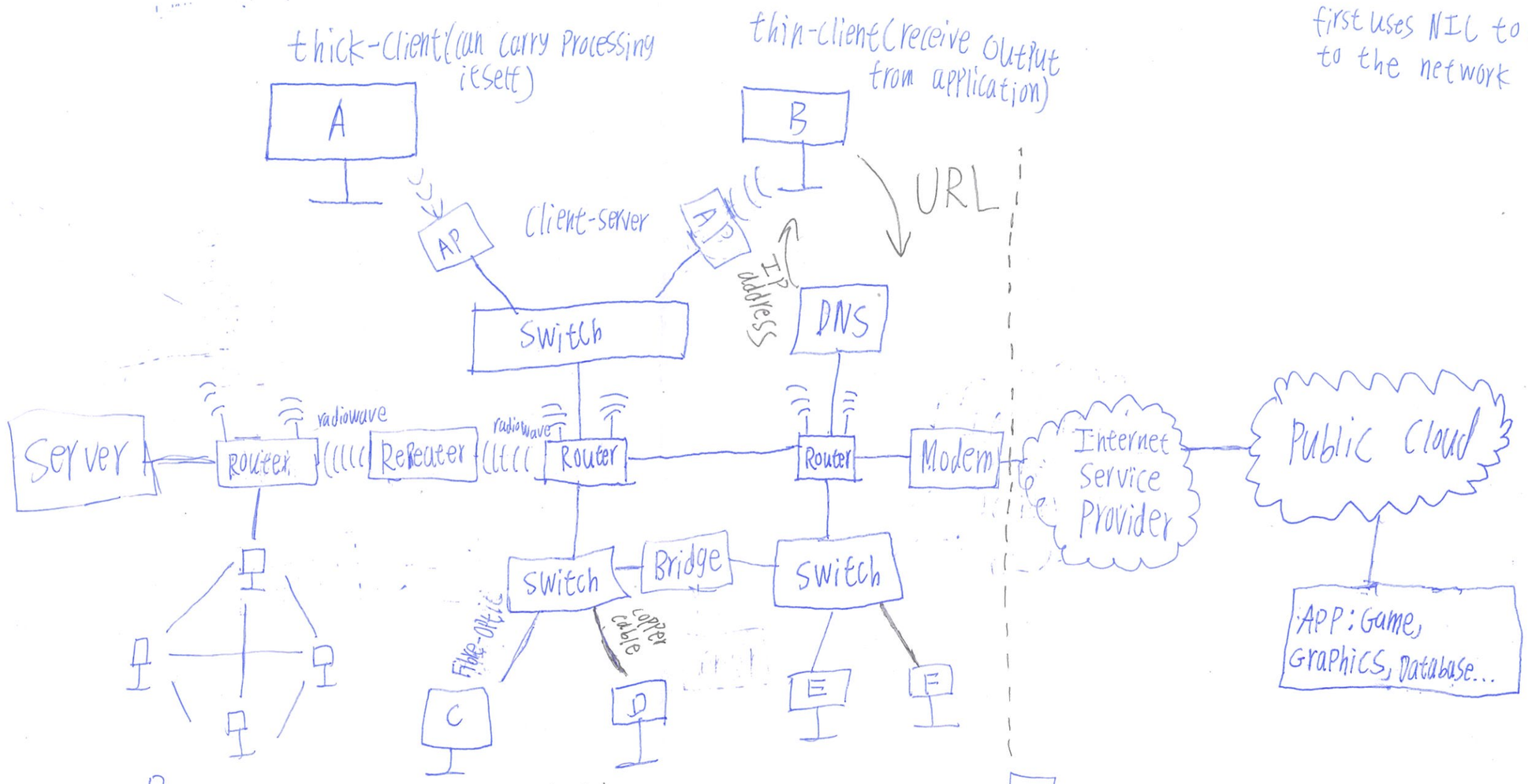


Cloud Computing



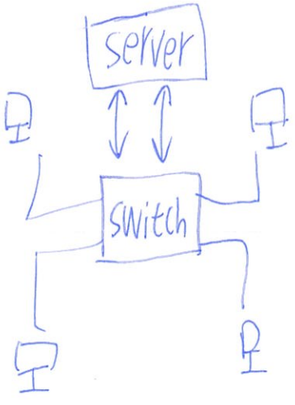
Yujie

- computers all first uses NIC to connect to the network

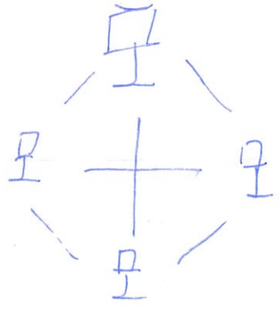


Peer-to-Peer

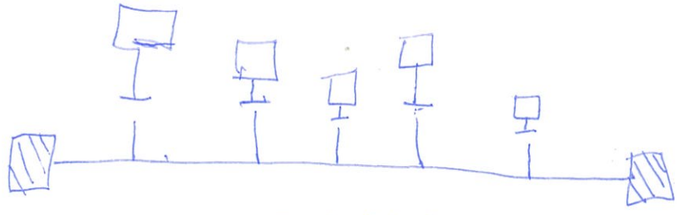
Hybrid network



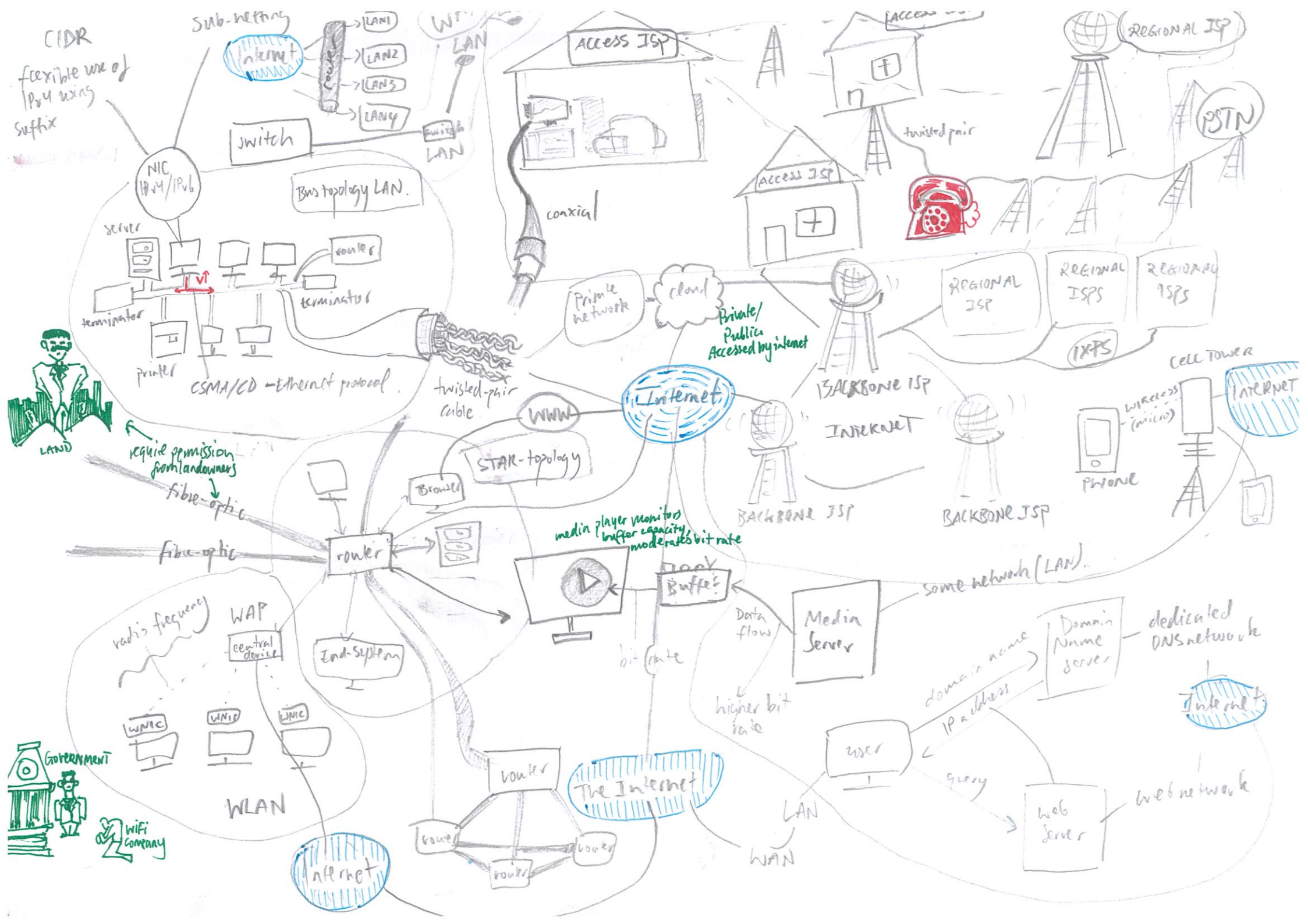
Star Topology



mesh topology



bus topology


CIDR
flexible use of IPv4 using suffix
Sub-netting
internet
router
LAN1
LAN2
LAN3
LAN4
switch
switch LAN
LAN
NIC IPv4/IPv6
Bus topology LAN.
server
router
terminator
terminator
printer
CSMA/CD - Ethernet protocol
twisted-pair cable
coaxial
Access ISP
Access ISP
twisted pair
REGIONAL ISP
PSTN
Private network
cloud
Private/ Public Accessed by internet
REGIONAL ISP
REGIONAL ISPs
REGIONAL ISPs
IXPs
BACKBONE ISP
INTERNET
BACKBONE ISP
BACKBONE ISP
CELL TOWER
wireless (micro)
INTERNET
PHONE
LAND
require permission from landowners
fibre-optic
fibre-optic
Internet
WWW
STAR-topology
Browser
router
media player monitors buffer capacity, moderates bit rate
Buffer
Data flow
Media Server
some network (LAN).
radio frequency
WAP
central device
End-System
WNIC
WNIC
LANIC
WLAN
bit rate
higher bit rate
Domain Name server
dedicated DNS network
domain name
IP address
Internet
user
query
web server
web network
GOVERNMENT
WiFi COMPANY
router
The Internet
router
router
router
LAN
WAN
Internet